

REPORT ON SENTIMENT ANALYSIS OF AIRLINE TWEETS USING TRADITIONAL AND DEEP LEARNING MODELS.

1. Definition

Natural Language Processing (NLP) allows computers to understand and interpret human language. It is a field of Artificial Intelligence. NLP tasks include text classification, translation, summarisation, and sentiment analysis, which require systems to go beyond keyword matching and capture meaning and context in text (Jurafsky and Martin, 2023).

Sentiment analysis focuses on classifying text according to expressed opinion, commonly as positive, neutral, or negative. It is widely applied to user-generated content, where large volumes of unstructured text make manual analysis impractical (Liu, 2020).

The NLP problem addressed in this project is the automatic classification of airline-related tweets into sentiment categories. Tweets are informal, and often ambiguous, which makes accurate sentiment classification difficult. Solving this problem requires NLP models that can process informal social media text and interpret context rather than relying on surface-level word patterns.

2. Scope

This project focuses on classifying the sentiment of airline-related tweets using Natural Language Processing techniques. Tweets are short, informal, and often unclear, which makes sentiment hard to measure and creates a challenging problem. This study compares traditional machine-learning and deep-learning models to examine how well different approaches handle noisy, real-world social media text.

By focusing on airline tweets, this study examines how modern language models perform on a practically relevant dataset where sentiment is often implied rather than explicitly stated. The scope is limited to English-language, sentence-level sentiment classification into positive, neutral, and negative categories. The project does not include topic modelling, multilingual analysis, or emotion detection beyond three sentiment categories.

3. Importance

Sentiment analysis of airline tweets is important both in practice and for NLP research. Airlines operate in a competitive service industry where customer satisfaction influences reputation and revenue. Studies show that analysing Twitter data helps airlines capture passenger opinions, identify service problems, and assess customer experience in ways that traditional surveys cannot provide (Md Saad et al., 2023; Badanik et al., 2023). As Twitter is widely used for real-time customer complaints about delays, cancellations, and service quality, automated sentiment analysis supports timely monitoring and response to emerging issues (Arnarsdóttir, 2017; Lutzky, 2024).

From an NLP perspective, airline tweets present a challenging domain due to informal language, and the frequent use of sarcasm, which remains difficult for sentiment models to handle reliably (Dhumpati et al., 2025). Although earlier studies have applied sentiment analysis to airline-related data, many focus on traditional machine-learning models or limited comparisons, without systematically evaluating modern deep learning and transformer-based approaches on the same dataset (Rajput, 2025). This project addresses this gap by directly comparing traditional classifiers, an LSTM model, and a BERT-based transformer on a real-world airline Twitter dataset.

4. Background Review

Sentiment analysis research has evolved from traditional machine-learning methods with sparse text features to deep learning and transformer-based approaches. This section reviews key NLP studies by outlining the methods used, their results, and their main strengths and limitations.

Aljedaani et al. (2022) studied sentiment analysis of US airline tweets using a hybrid approach combining TextBlob with deep learning models, including LSTM, CNN and LSTM–GRU network. Their results showed that the LSTM–GRU outperformed other models with 97% accuracy and 0.96 for F1-score. The main strength of this work is the integration of dictionary-based sentiment information with neural models. However, the approach relies on pre-defined sentiment dictionaries and does not evaluate transformer-based architectures, limiting its ability to capture richer contextual meaning.

Liu (2020) reviewed earlier sentiment analysis methods, including Naive Bayes, Support Vector Machines, and simple neural models. The review found that classical approaches using bag-of-words or TF–IDF features are efficient, interpretable, and suitable for small or imbalanced datasets, but struggle to capture word order, negation, and context, reducing effectiveness on informal social media text.

With the introduction of transformers, Devlin et al. (2019) proposed BERT, which uses bidirectional self-attention to capture context from both directions. In related work, Murfi et al. (2024) combined BERT with CNN and LSTM layers and achieved higher accuracy and F1-score than traditional and recurrent models. While effective, their study highlights the high computational cost and careful fine-tuning required for BERT-based models.

Khan et al. (2022) applied BERT-based contextual embeddings to sentiment analysis of noisy user-generated text using neural classifiers such as CNN. Their results showed improved performance over feature-based methods on informal text. However, the study does not provide a broad comparison with traditional or recurrent models or examine class imbalance in detail.

Chaudhary et al. (2023) reviewed deep learning approaches for Twitter sentiment analysis, covering CNN, LSTM, BiLSTM, hybrid architectures, and transformer-based models. They reported that deep learning methods generally outperform traditional machine-learning models on Twitter data, while also identifying ongoing challenges such as sarcasm, domain shift, class imbalance, and the computational cost of large transformers.

Overall, these studies show trade-offs between efficiency, interpretability, and how well models can capture context across traditional, recurrent, and transformer-based sentiment analysis methods.

5. SMART Objective

The goal of this project is to develop and evaluate NLP models for sentence-level sentiment analysis of airline customer tweets using the Twitter US Airline Sentiment dataset from Kaggle. The task is to classify tweets as positive, neutral, or negative to assess passenger sentiment toward major US airlines.

Two traditional machine-learning models (Naive Bayes and Support Vector Machine) and two deep learning models (LSTM and BERT) will be implemented and compared. Traditional models will use TF-IDF features, while deep learning and transformer-based models will use learned embeddings. Performance will be measured using accuracy, precision, recall, and F1-score, with particular attention to the neutral class, which is known to be difficult to classify in social media text. The project aims to achieve a minimum test accuracy of 80% for the best-performing model.

The objective is attainable, as the dataset is publicly available, pre-labelled, and widely used in prior sentiment analysis studies. Existing NLP research shows that LSTM and BERT-based models perform well on noisy social media text, supporting the feasibility of the proposed methods and performance target (Acheampong et al., 2021; Devlin et al., 2019).

The objective is relevant because analysing airline tweets provides actionable insight into customer satisfaction and service quality, contributing to practical applications of NLP in customer experience analysis (Md Saad et al., 2023; Liu, 2020). The project is time-bound by the module assessment schedule and will be completed before the assignment submission deadline.

6. Dataset Description

This project uses the Twitter US Airline Sentiment dataset, publicly available on Kaggle. The dataset contains English-language tweets directed at major US airlines, each annotated with a sentiment label. The dataset is text-based and tabular, with each row representing a single tweet. For this study, only the tweet text and sentiment label were used, as other metadata fields were not required for sentiment classification.

The dataset contains 14,640 tweets classified into positive, neutral, and negative sentiment. The class distribution is imbalanced, with negative tweets forming the majority class, reflecting typical customer behaviour on social media. To address this, model performance is assessed using precision, recall, and F1-score in addition to accuracy.

Overall, the dataset is suitable for NLP-based sentiment analysis due to its real-world social media text, pre-labelled structure, and challenging class imbalance, making it appropriate for evaluating both traditional and deep learning models.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was carried out to understand the Twitter US Airline Sentiment dataset and to guide model design and evaluation. The analysis focused on text preprocessing, class distribution, text characteristics, and establishing a baseline.

7.1. Data Cleaning and Preprocessing

All tweets were converted to lowercase for consistency. URLs, user mentions, hashtags, numbers, and punctuation were removed to reduce noise, and extra whitespace was eliminated. Tweets were tokenised using spaCy, after which stopwords were removed and lemmatisation was applied to reduce vocabulary size and group related word forms. These steps support better generalisation, particularly for traditional machine-learning models.

7.2. Class Distribution and Imbalance Handling

Analysis of sentiment labels revealed clear class imbalance, with negative tweets forming the majority class (Figure 7.1). This imbalance can bias models toward the dominant class and reduce minority-class performance. To address this, class-sensitive training was applied: class priors for Naive Bayes and class weights for the Support Vector Machine, LSTM, and BERT models, preserving the original data distribution.

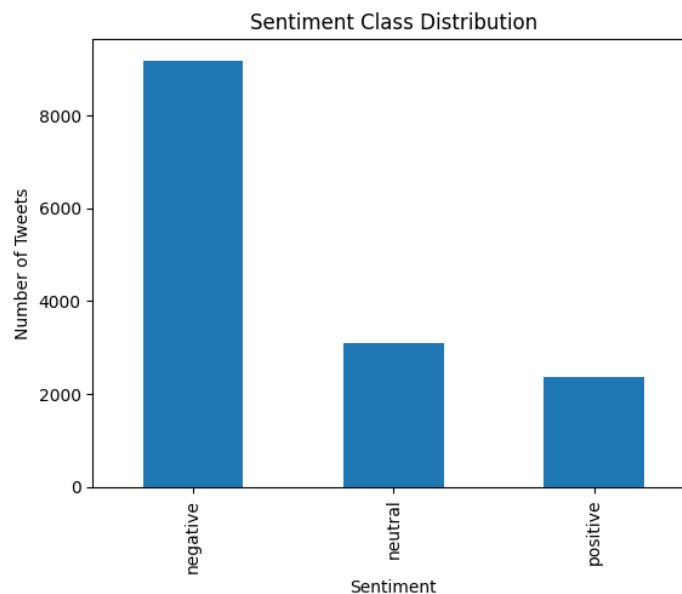
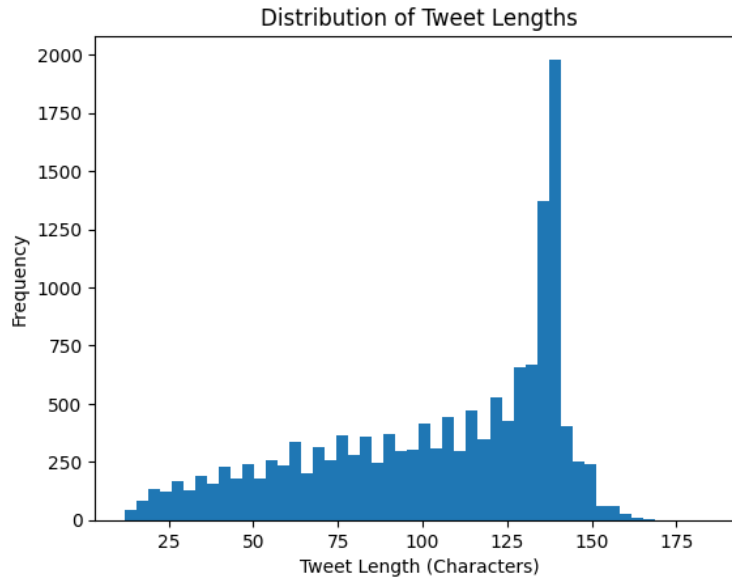


Fig 7.1. Distribution of the three sentiment classes

- **7.3. Text Length Analysis**

A histogram of tweet lengths shows that most tweets fall within a moderate length range, with many close to Twitter's character limit (Figure 7.2). This supports using a fixed sequence length, as most tweets can be represented without excessive truncation.



7.4. Word Cloud Analysis

A word cloud of the cleaned text highlights frequent terms such as *flight*, *delay*, *help*, and *time* (Figure 7.3), indicating that tweets largely focus on service issues and disruptions. This confirms the dataset's relevance to airline sentiment analysis.



7.5. Vectorisation

Text was converted into numerical representations using TF-IDF features for traditional models, integer tokenisation with trainable embeddings for the LSTM, and subword token embeddings generated using the BERT tokenizer.

7.6. Baseline Model

A majority-class baseline was used, in which all tweets are predicted as negative. This baseline achieves approximately 62.7% accuracy but yields zero recall and F1-score for neutral and positive classes, providing a lower-bound benchmark for evaluating all models.

8. Traditional Machine Learning Methods

Traditional machine learning methods are techniques commonly applied to sentiment analysis tasks and typically rely on sparse text features such as bag-of-words or TF-IDF representations. They are valued for their simplicity, efficiency, and interpretability. This section reviews five relevant models and explains the selection of two methods for this project.

8.1. Naive Bayes (NB)

Naive Bayes is a probabilistic model that treats each feature as independent from the others. Despite this simplification, it performs well on high-dimensional sparse text and trains very quickly, making it a strong baseline, particularly for small datasets (Dou et al., 2023). However, its assumption of independence limits its ability to capture word relationships and context, reducing performance on informal text such as tweets (Liu, 2020).

8.2. Support Vector Machine (SVM)

SVM is a widely used supervised model for sentiment analysis. When combined with TF-IDF features and a linear kernel, it performs well on high-dimensional text by learning decision boundaries that maximise class separation (Jurafsky and Martin, 2023). A key limitation is sensitivity to class imbalance, which often requires careful tuning of the regularisation parameter and the use of class weighting to achieve balanced performance (Rozi et al., 2025; Olawale, Iworiso and Amaunam, 2023).

8.3. Logistic Regression

Logistic Regression is a simple linear classification method commonly used as a baseline in sentiment analysis. It is efficient and interpretable but is limited to linear decision boundaries, making it less suitable for modelling complex sentiment patterns without feature engineering (Jurafsky and Martin, 2023).

8.4. Decision Trees

Decision Trees are interpretable and can learn non-linear patterns, but they generally perform poorly on sparse text features and are prone to overfitting when applied to noisy sentiment data (Sruti and Shreyas, 2024).

8.5. K-Means Clustering

K-Means Clustering is an unsupervised learning approach used for exploratory sentiment grouping. While useful for identifying latent structures, it lacks sentiment labels and typically underperforms supervised classifiers when annotated data is available (Patel and Goswami, 2024).

Based on this comparison, Naive Bayes and Support Vector Machine were picked for this project. Both are suitable to TF-IDF features and provide efficient, widely used benchmarks. Naive Bayes offers a simple probabilistic baseline, while SVM provides stronger classification performance, making them appropriate for comparison with deep learning models.

9. Deep Learning Methods

Deep learning methods are commonly used in sentiment analysis because they can learn features directly from text and capture context more effectively than traditional models, particularly on noisy social media data (Chaudhary et al., 2023).

9.1. Convolutional Neural Networks (CNNs)

CNNs use convolutional layers on word embeddings to identify local word patterns. They are efficient and perform well on short texts such as tweets. However, CNNs mainly focus on local context and struggle to learn long-range dependencies within a sentence.

9.2. Recurrent Neural Networks (RNNs)

RNNs handle text one token at a time and learn patterns across sequences. While they were an early deep learning approach for sentiment analysis, standard RNNs suffer from vanishing gradient problems and are ineffective at capturing long-term dependencies, which has limited their use in recent work (Tan et al., 2023).

9.3. Long Short-Term Memory networks (LSTMs)

LSTM addresses these limitations through gating mechanisms that control information flow. LSTMs have shown strong performance on sentiment analysis tasks by capturing sequential patterns and longer-range relationships in text more effectively than basic RNNs (Chaudhary et al., 2023; Derbentsev et al., 2023). Their main drawback is higher computational cost and a greater risk of overfitting on smaller datasets.

9.4. Bidirectional LSTMs (BiLSTMs)

BiLSTMs extend LSTMs by analysing text from both directions of a sequence, allowing the model to incorporate wider contextual information. This often improves performance on short texts but increases training time and computational requirements (Chaudhary et al., 2023).

9.5. Bidirectional Encoder Representations from Transformers (BERT)

BERT uses self-attention to learn context from both the left and right sides of a sentence simultaneously. BERT has achieved strong results on sentiment analysis benchmarks and is suitable for informal social media text, as it captures subtle and context-dependent sentiment signals (Acheampong, Nunoo-Mensah and Chen, 2021; Chaudhary et al., 2023). However, it requires significant computational resources and careful fine-tuning.

Based on this comparison, LSTM and BERT were selected for this project. LSTM provides an effective sequence-based model that can be trained on moderate hardware, while BERT offers

state-of-the-art contextual modelling and strong performance on social media sentiment analysis when fine-tuned appropriately.

10. Implementation and Refinement

All models were developed in Python using standard NLP and machine-learning libraries, including scikit-learn, spaCy, TensorFlow/Keras, and Hugging Face Transformers. Each pipeline was trained, tuned, and evaluated using accuracy, precision, recall, and F1-score.

10.1. Naive Bayes

Naive Bayes was implemented using scikit-learn with text preprocessing carried out using spaCy and data handling using Pandas and NumPy. Tweets were represented using TF-IDF features with unigrams and bigrams, fitted on the training data only, to prevent information leakage.

Two refinement strategies were applied:

- **Hyperparameter tuning** of the smoothing parameter (*alpha*) using GridSearchCV to improve generalisation.
- **Class priors** were applied to reduce bias toward the majority negative class.

10.2. Support Vector Machine (SVM)

SVM was implemented using scikit-learn, following the same preprocessing and TF-IDF representation as Naive Bayes. A linear SVM was trained and evaluated using standard classification metrics.

Refinement strategies included:

- **Hyperparameter tuning** of the regularisation parameter (*C*) using GridSearchCV with cross-validation.
- **Class weighting** to penalise misclassification of minority sentiment classes and improve balanced performance.

10.3. Long Short-Term Memory (LSTM)

The LSTM model was implemented using TensorFlow and Keras. Tweets were tokenised into integer sequences, padded to a specific length. The input was passed through a trainable embedding layer, an LSTM layer, and a softmax output, with the data divided into training, validation, and test sets.

Refinement strategies included:

- **Hyperparameter tuning**, adjusting LSTM units, batch size, dropout rate, number of epochs and learning rate to improve generalisation.
- **Regularisation and class-sensitive training**, using dropout, early stopping based on validation loss to reduce overfitting and class weights, to improve minority-class performance.

10.4. BERT

The BERT model was implemented using TensorFlow and the Hugging Face Transformers library. Tweets were tokenised using the bert-base-uncased tokenizer, producing token IDs and attention masks. Dataset splitting and evaluation followed the same procedure as the LSTM model.

Two refinement strategies applied:

- **Transfer learning and fine-tuning**, starting with frozen encoder layers and then fully fine-tuning all layers using a low learning rate, to adapt BERT to airline tweet sentiment.
- **Class weighting** during training to reduce bias toward the majority class.

11. Evaluation

All models were evaluated using accuracy, precision, recall, and F1-score, with attention to class-level performance. Tables and confusion matrices summarise results and illustrate the effect of model refinement.

11.1. Naive Bayes

Naive Bayes was evaluated using default and tuned configurations. The default model achieved 0.74 accuracy but showed bias toward the majority negative class, resulting in low macro F1-score of 0.59 (Table 11.1).

Table 11.1. Performance summary of Naive Bayes models

Model Variant	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
Naive Bayes (Default)	0.74	0.78	0.55	0.59
Naive Bayes (Tuned)	0.78	0.78	0.65	0.69

After tuning the smoothing parameter, performance improved to 0.78 accuracy and 0.69 macro F1-score. Confusion matrix (Figure 11.2) shows reduced misclassification of neutral and positive tweets compared to Figure 11.1. These results indicate that while Naive Bayes is efficient, refinement is necessary to mitigate class imbalance.

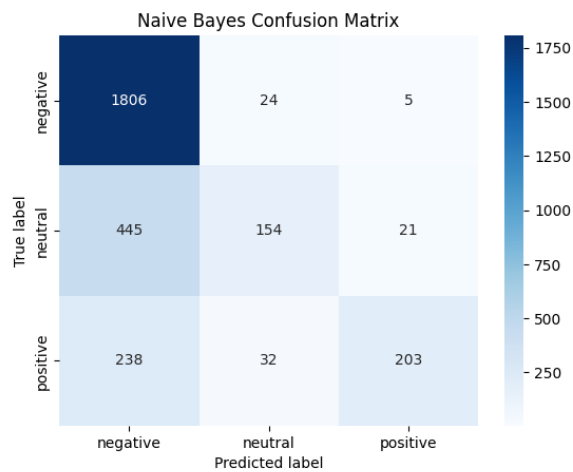


Fig 11.1. Confusion Matrix for default Naïve Bayes

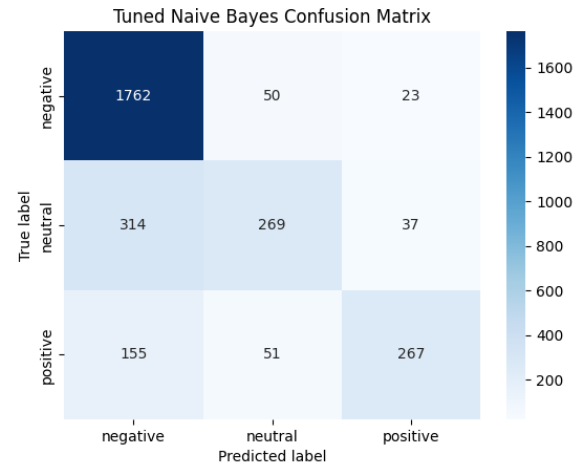


Fig 11.2. Confusion Matrix for fine-tuned Naïve Bayes

11.2. Support Vector Machine (SVM)

The default SVM achieved 0.79 accuracy and a macro F1-score of 0.74 (Table 11.2), showing more balanced performance than Naive Bayes. Some confusion between neutral and negative tweets remained (Figure 11.3).

Table 11.2. Performance summary of Support Vector Machine

Model Variant	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
SVM (Default)	0.79	0.74	0.73	0.74
SVM (Tuned)	0.81	0.77	0.74	0.75

After tuning the regularisation parameter and applying class weights, accuracy increased to 0.81 with a macro F1-score of 0.75. Figure 11.4 shows improved class separation, particularly for neutral tweets. Overall, SVM demonstrated robustness on high-dimensional text and benefited from targeted refinement.

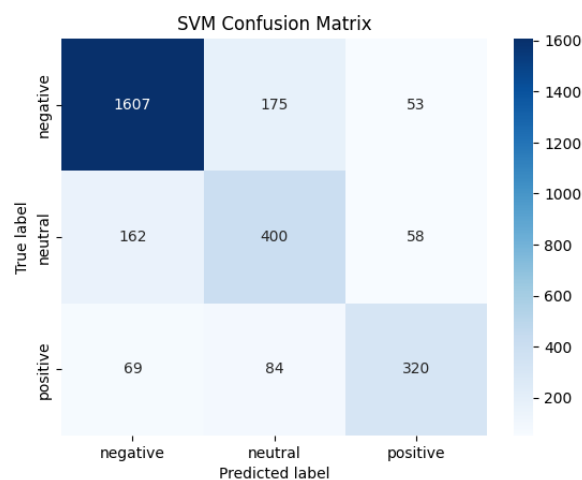


Fig 11.3. Confusion Matrix for default SVM

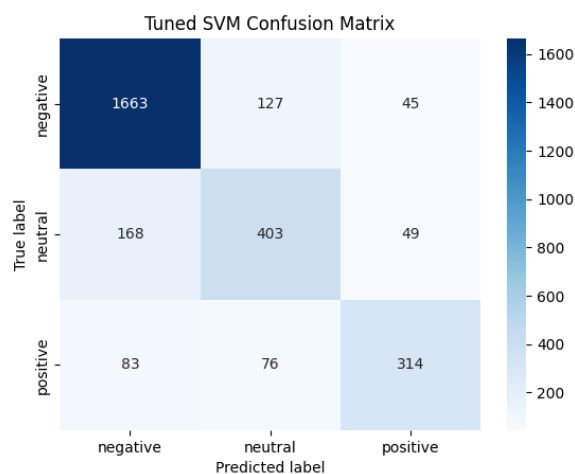


Fig 11.4. Confusion Matrix for fine-tuned SVM

11.3. Long Short-Term Memory (LSTM)

The default LSTM showed increasing training accuracy but unstable validation performance (Figures 11.5 and 11.6), indicating overfitting. This resulted in a macro F1-score of 0.70 (Table 11.3).

Table 11.3: Performance Comparison of LSTM Variants

Model Variant	LSTM Units	Epoch	Dropout	Early Stopping	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
LSTM (Default)	[64]	10	0.3	False	0.76	0.69	0.70	0.70
LSTM (Tuned 1)	[64]	10	0.5	True	0.77	0.71	0.73	0.72
LSTM (Tuned 2)	[64, 128]	15	0.3	True	0.78	0.73	0.72	0.72
LSTM (Tuned 3)	[64, 128]	10	0.5	False	0.76	0.70	0.69	0.70

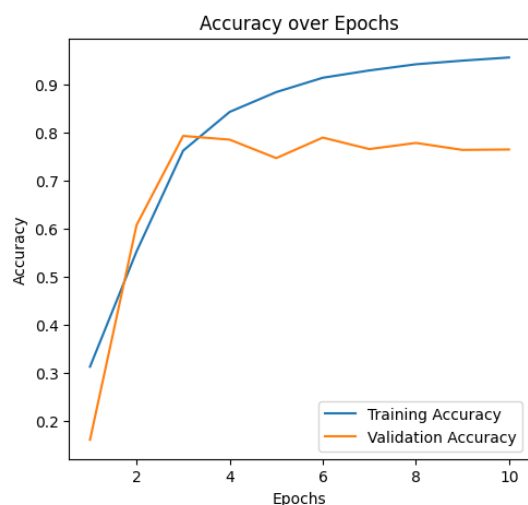


Fig 11.5. Training and validation accuracy of default LSTM

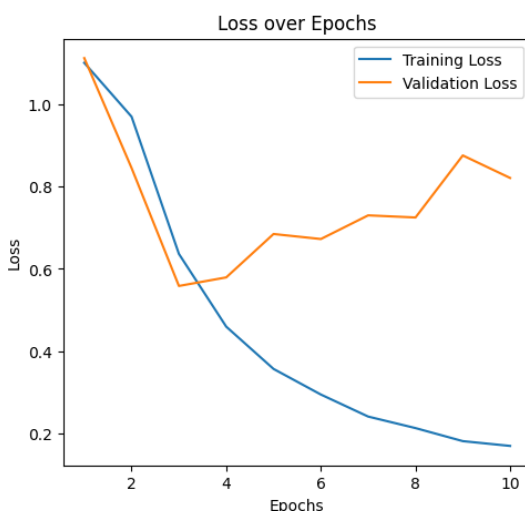


Fig 11.6. Training and validation loss of default LSTM

The refined LSTM achieved 0.78 accuracy and a macro F1-score of 0.72. Training curves (Figures 11.7 and 11.8) show improved stability due to hyperparameter tuning and early stopping.

These results demonstrate that LSTMs can capture sequential patterns in tweets but are sensitive to architecture and training configuration.

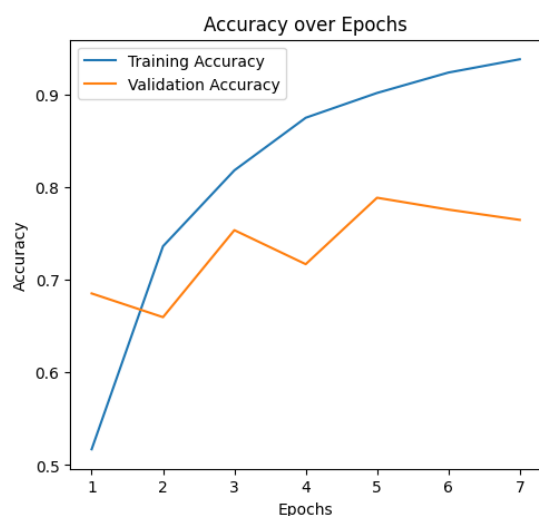


Fig 11.7. Training and validation accuracy of best

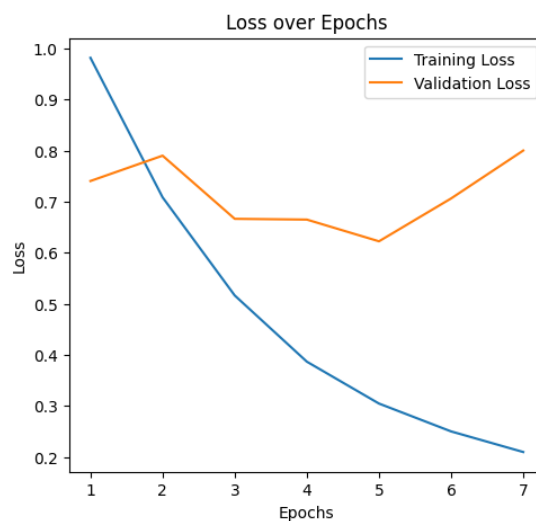


Fig 11.8. Training and validation loss of best LSTM

11.4. Bidirectional Encoder Representations from Transformers (BERT)

The frozen-encoder BERT baseline underperformed, achieving 0.59 accuracy and a macro F1-score of 0.49 (Table 11.4), reflecting underfitting (Figures 11.9 and 11.10).

Table 11.4: Performance Comparison of BERT Variants

Model Variant	Frozen Encoder	Epochs	Learning Rate	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
BERT (Default)	True	2	2e-5	0.59	0.49	0.50	0.49
BERT (Tuned 1)	False	2	2e-6	0.78	0.72	0.78	0.74
BERT (Tuned 2)	False	4	2e-5	0.83	0.78	0.77	0.77

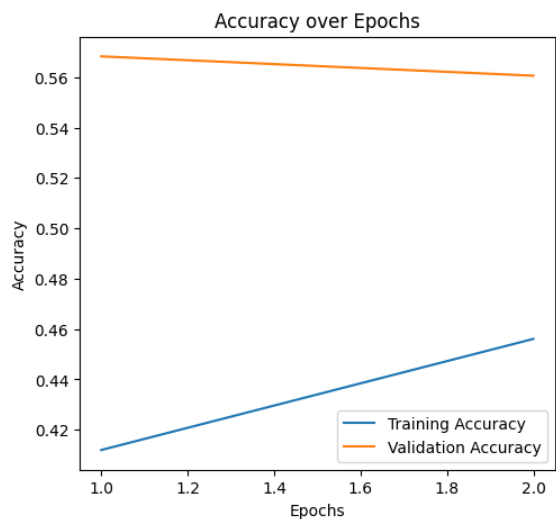


Fig 11.9. Training and validation accuracy of default BERT

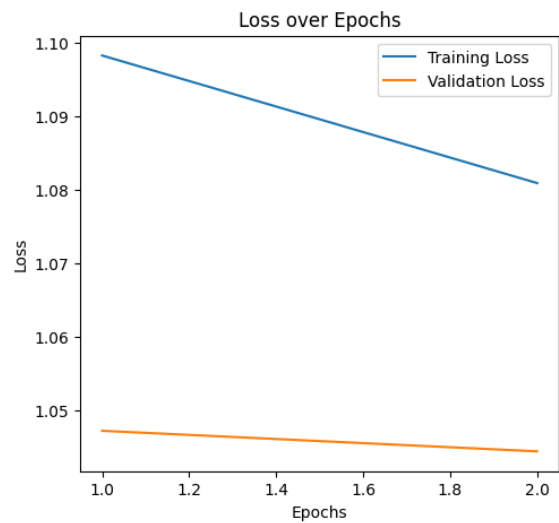


Fig 11.10. Training and validation loss of default BERT

Fine-tuning the encoder significantly improved performance. The first fine-tuned model achieved 0.78 accuracy and 0.74 macro F1-score, while the best configuration reached 0.83 accuracy and 0.77 for macro F1-score.

Training curves show rapid convergence with stable early validation performance (Figures 11.11 and 11.12). These results confirm that full fine-tuning enables BERT to capture contextual and ambiguous sentiment more effectively than other models.

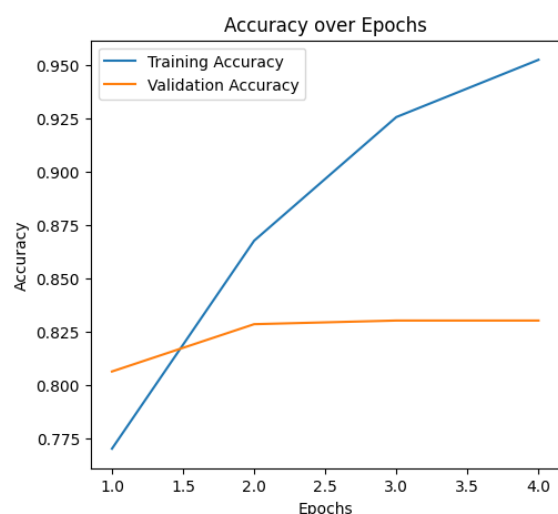


Fig 11.11. Training and validation accuracy of best BERT

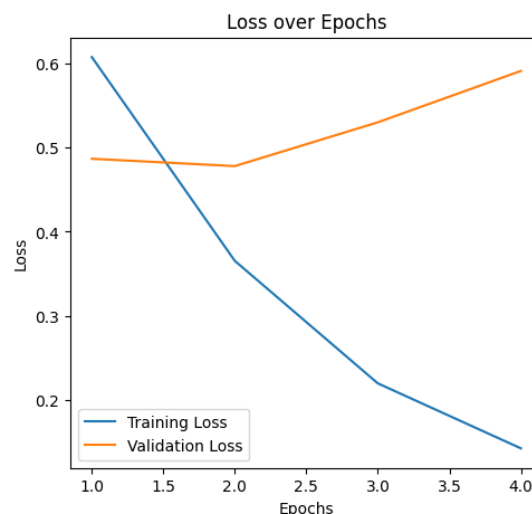


Fig 11.12. Training and validation loss of best BERT

12. Conclusion

This project evaluated sentiment analysis of airline-related tweets using four NLP models: Naive Bayes, Support Vector Machine, LSTM, and BERT, applied to the Twitter US Airline Sentiment data. Performance was assessed using accuracy, precision, recall, and F1-score, focusing more on the neutral class, which is difficult to model in social media text.

The results show clear differences between modelling approaches. Naive Bayes provided a simple and efficient baseline but required explicit handling of class imbalance to achieve balanced results. Support Vector Machine consistently delivered strong and stable performance when combined with TF-IDF features and class weighting. The LSTM model captured sequential patterns in tweets but was sensitive to architectural and training choices. The fine-tuned BERT model achieved the best overall performance, reaching 0.83 accuracy and the highest macro F1-score, confirming the strength of transformer-based models for capturing contextual and ambiguous sentiment in noisy text.

Overall, the findings highlight the importance of model refinement and careful evaluation when applying NLP methods to real-world sentiment analysis tasks. The structured comparison illustrates trade-offs between simplicity, interpretability, computational cost, and performance, and shows that transformer-based models are particularly well suited to analysing airline customer feedback on social media.

Despite strong performance, challenges remain. Class imbalance continues to affect neutral and positive sentiment prediction, and computational constraints limit extensive tuning of transformer models. Future work could explore larger datasets, advanced imbalance-handling techniques, and further optimisation of deep learning architectures.

This report uses the Harvard reference style, which applies author-date in-text citations with an alphabetically ordered reference list. MLA is commonly used in the humanities, APA in the social

sciences, and Vancouver uses numerical references mainly in medical research. Chicago supports both author–date and footnote systems but can increase formatting complexity. Harvard was selected for its clarity and suitability for technical NLP reports.

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